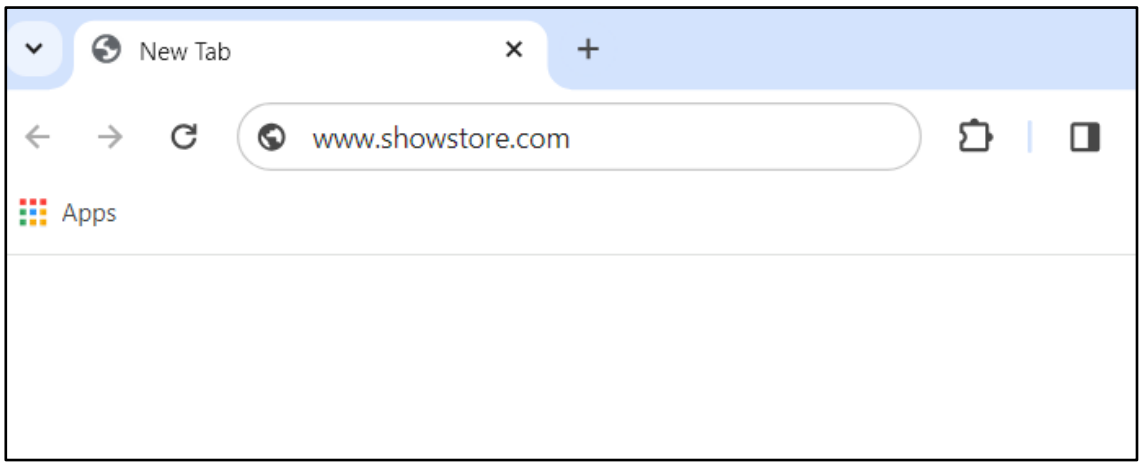
How Websites Work?

1. **User Input and Request:**

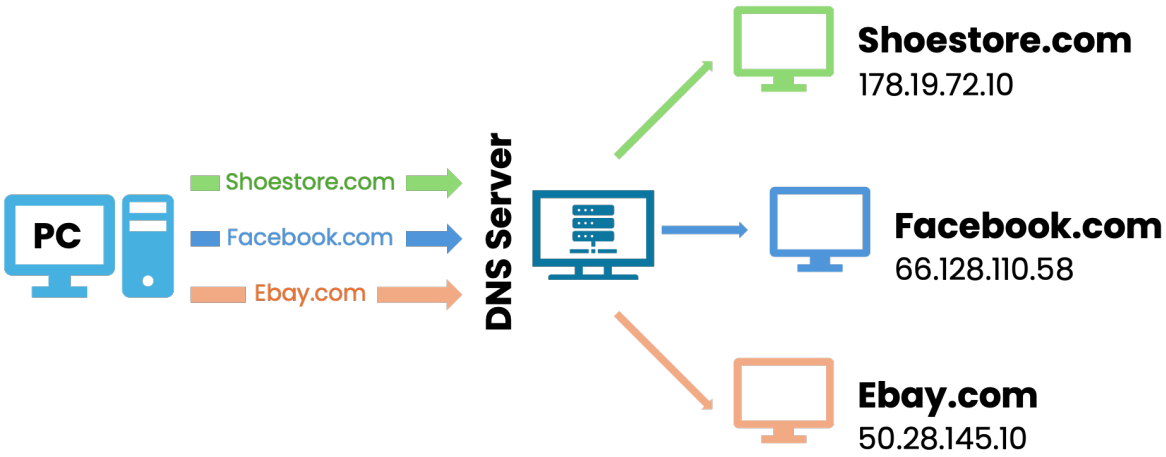
   Open a web browser (like Chrome, Firefox, Safari) and enter a website's URL into the address bar like **www. *shoestore. com***. This is the domain name of the website, which is a unique identifier for the website address on the internet.



Alt text: *Website address in the address bar*

2. **Domain Name System (DNS) Resolution:**

   Your web browser sends a request to a DNS server to translate the domain name into an IP address. An IP address is a numerical address that identifies a specific computer on the internet. The IP address helps in identifying and routing data to and from devices on the internet.



Alt text: *Conversion of website address into a numeric address*

DNS server is responsible for storing the official records of the domain. DNS server provides the IP address of the server to your computer. DNS Server functions like a translator, that turns easy-to-remember domain names into the required numerical IP addresses for routing data across the Internet. This way, users can access websites using familiar names rather than dealing with complicated numerical IP addresses.

3. **HTTP Request:**
   An HTTP (Hypertext Transfer Protocol) request is a message sent by a client (typically a web browser) to a server, requesting a particular action to be performed. HTTP request is a method for clients to communicate with servers, requesting resources or actions. Once the browser has the server's IP address, it sends an HTTP (Hypertext Transfer Protocol) request to the server, asking for the website's data.

Alt text: *Http request to web server*

4. **Server Processing:**

   The server receives the request and processes it. This involves various actions, including retrieving data from a database, handling business logic, and preparing the requested content.

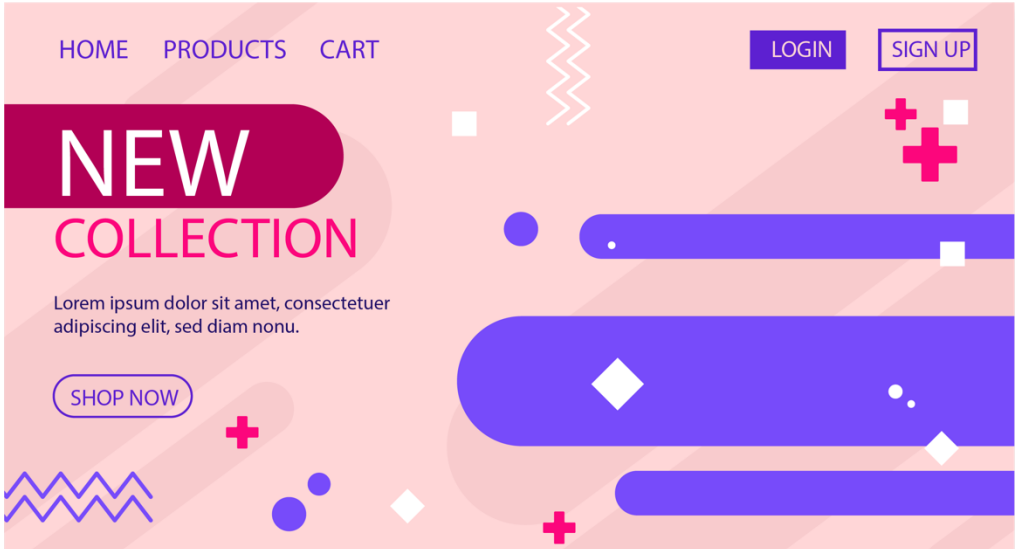
5. **Server Response:**

Alt text: *Response to browser*

After processing the request, the server sends back an HTTP response to the user's browser. This response contains the requested data, usually in the form of HTML, CSS, and JavaScript files.

6. **Browser Rendering:**

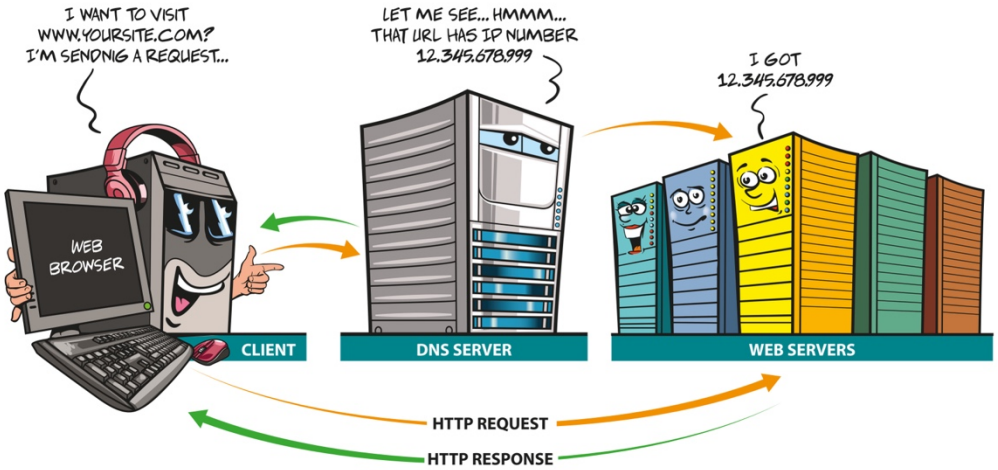
The browser receives the response and begins rendering the web page. It interprets the HTML to structure the content, applies styles with CSS, and executes JavaScript to add interactivity.



Alt text: *Rendered website*

7. **Website on Screen:**

You can now see the website on your screen.
The provided image illustrates the entire website rendering process as discussed.



Alt text: *Working of websites*